Effectivity

The following CT Equipment is affected:

2266832	LightSpeed Plus Console Assembly
2266832-2	LightSpeed Plus Console Assembly w/D3D

Purpose

The purpose of this FMI is to simplify the Console SCSI chain, by eliminating the external SCSI/ serial expander. Completion of this FMI will also prepare the console for a future upgrade to the Pegasus Image Generator board.

Related FMIs

- FMI25276: Pegasus IG Upgrade
- FMI25277: H2 M2.1 Software Upgrade

Furnished Materials

FMI Kit Part Number 2296263 includes the items listed below.

Qty	Description	Part #
1	FMI Document	2289363-100
1	High Density SCSI card	2169940-9
1	Serial card (Digi)	2282000
1	Intercom/Interconnect octopus	2269875-2
1	Serial octopus	2269874-2
1	SCSI to DASM cable	2277403
1	SCSI cable	2277402
1	Nut Driver (Stand-off Tool) Amp P/N 811262-1	2264575
AR	Ty-Raps	

Special Tools and Test Equipment

Qty	Description	Part #
1	Phillips Head Screwdriver	
1	3/32” or 5mm socket	
1	ESD Wrist Strap	

Qty	Description	Part #
1	4mm Hex Wrench (for front cover)	
1	Diagonal cutters (for ty-raps)	
1	Compression Cap (recommended)	2169940-21

Disposal & Removed Parts

Destroy and/or dispose of all obsolete hardware, software and documentation according to GE policy. The following parts will be removed from the console, for this FMI, and will *not* be reused:

Qty	Description	Part #
2	PCI Module filler plates	
1	SCSI Cable, Octane-DASM	2191432-2
1	SCSI 4-Port Expander, incl. PS & PS Cable	2174053
1	Bracket - Serial Port Expander	2173323
1	Bracket - AC Adapter	2173330
1	SCSI Cable, Central Data	2191439-2
1	Cable - Octopus	2269874
1	Cable - Side bulkhead to RIP Bd.	2269875
	Assorted Fasteners	

People Power

Allow 2 hours of labor for the completion of this SCSI chain upgrade. Allow an additional 2 hours for loading new software.

Procedure

NOTICE Potential for Data Loss

Save System State before proceeding, particularly if this FMI is being performed in conjunction with FMI25276 (Pegasus IG Upgrade) or FMI25277 (H2 M2.1 Software Upgrade).

DO NOT load software until after both this FMI and FMI25276 are completed (if applicable).

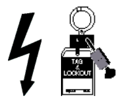
1 Preparation

Purpose: The purpose of this section is to 1) Ensure that no circuits are live within the Console and 2) Remove the console covers to gain access to the components therein.

1.) Bring software down.
2.) Shutdown system, via normal shutdown procedures.

3.) Tag and lockout the main system power, supplied to the PDU (i.e., main disconnect).

WARNING



POTENTIAL FOR PERSONAL INJURY OR DEATH BY ELECTRIC SHOCK. TO AVOID RISK OF ELECTRIC SHOCK, FOLLOW PROPER LOCKOUT/TAGOUT PROCEDURES.

4.) Disconnect J19, J20 and J52 from the rear of the console. Remove the back panel.

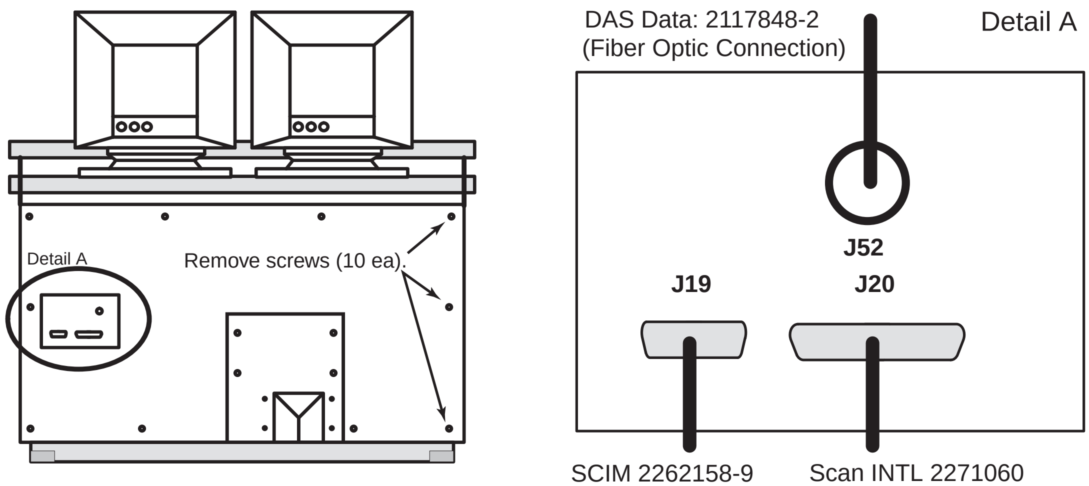


Figure 1 Rear of Console, showing bulkhead and screw locations

5.) Remove the console's front cover, using the 4mm hex wrench.

6.) Pull out the platform upon which the Octane computer rests. Release its tie-down strap, if present.

7.) Disconnect the SCSI, Line-In (R), Line-In (L) and headphone cables from the rear of the Octane. Remove any ty-raps that bundle these cables to others, as necessary.

2 PCI Expansion Card Installation

NOTICE

Wear a grounded ESD wrist strap. Place removed electronic parts on an anti-static surface.

1.) Remove the Ethernet RJ45 cables from the Octane's PCI module.

2.) Loosen the two captive phillips screws (see Figure 2).

3.) Pull out the release lever along the bottom of the module (see Figure 2).

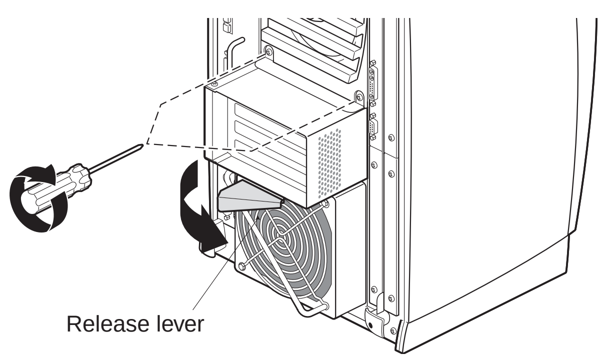


Figure 2 Releasing the PCI Module

4.) Slide out the module, taking care not to allow the compression connector to touch anything. Place a protective cap (2169940-21) on the compression connector.

NOTICE
Avoid Damage

Never touch the gold surface of the compression connector. Touching it could damage the connector. Place a protective cap on the connector to prevent damage.

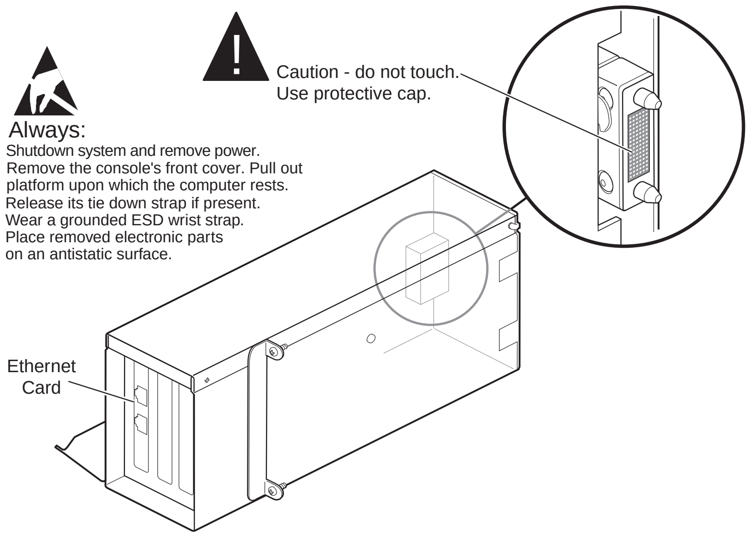


Figure 3 Placing the PCI Module for Board Installation

5.) Lie the module on its right side and loosen the two phillips screws that hold the left side access cover (see Figure 4). Then lift and remove this cover.

6.) Remove two (2) filler plates (these will not be reused). Retain the mounting screws for reuse.

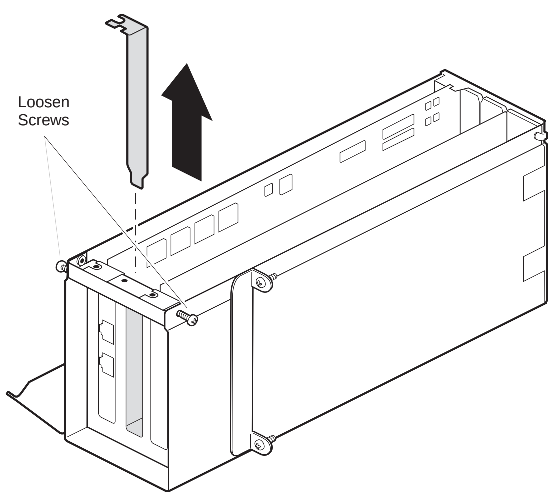


Figure 4 Removing the panel for a new card

NOTICE **PCI cards are extremely sensitive to static electricity.**

7.) Insert the serial card (2282000) in the middle slot, above the ethernet card (already in place).

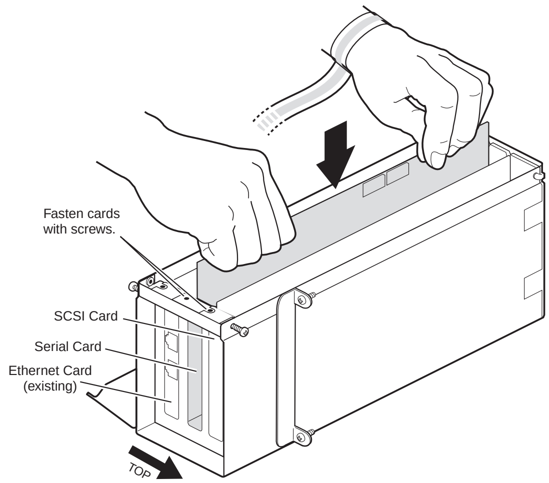


Figure 5 Installing a board into the PCI Module

8.) Connect the serial connector of the new serial octopus (2269874-2) to the serial card (see Figure 6). When the card is properly centered, finger tighten the thumbscrews of the connector first, and *then* (and only then) fasten the card in place with a screw (see Figure 5).

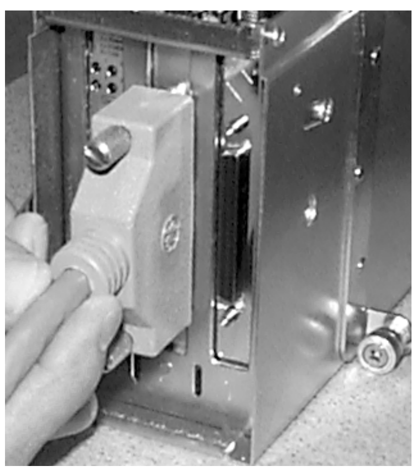


Figure 6 Use the Serial Connector to properly center the Serial Card

9.) Install the high density SCSI card (2169940-9) in the top slot, above the serial card.
- Center the card in the slot, so that the I/O cable will fit correctly (the cable does not need to be attached at this time).
- Fasten the card in place with a screw (see Figure 5).

10.) To reinstall the PCI module, reverse Steps 1 through 5 (above). Make certain to:
- Remove protective compression cap from the rear of the module.
- Reconnect the Ethernet RJ45 cables.

11.) Connect the remaining three cables of the serial octopus (2269874-2) to the Line-In (L and R) and headphone ports of the Octane.

Ty-rap the new cables to the handle, to avoid their being pulled out inadvertently.

3 Cable Replacement

Note: Refer to the block diagram shown in Figure 13, on page 9, for proper cable routing.

1.) Disconnect the two (2) SCSI cables and three (3) RJ45 cables from the SCSI/serial expander (Central Data box). Then remove the data box, power supply, associated hardware and power supply cable. The freed SCSI cable may be discarded.

- Remove ty-raps as necessary.
- The top mounting bracket may be left in place (removal is not necessary).
- Refer to Figure 8 for location of SCSI/serial expander, and Figure 7 for location of screws.

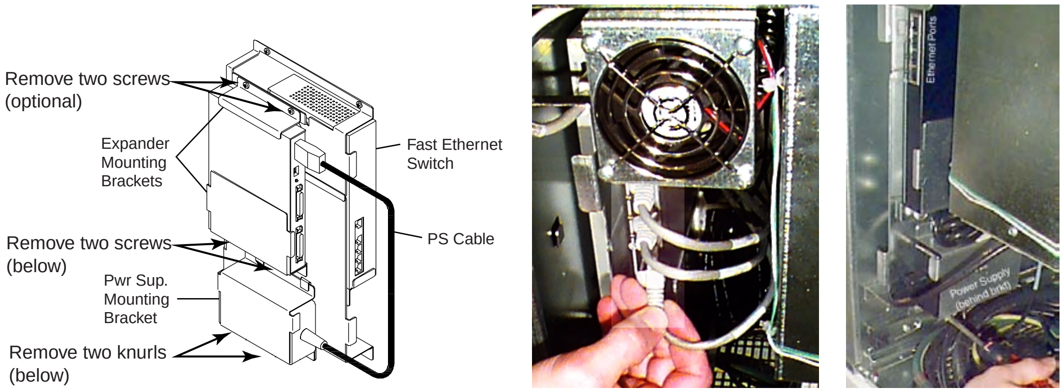


Figure 7 Central Data serial expander

2.) Remove the front EMC cover from the “ICEbox”, using a flat-blade screwdriver.

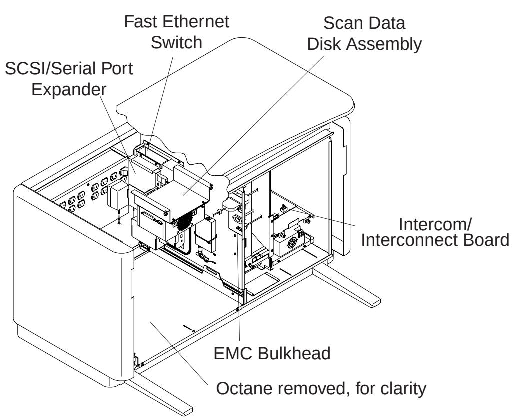


Figure 8 Console, with front cover and EMC cover removed

3.) Disconnect the SCSI cable at J23, from the Octane side of the side bulkhead (see Figure 8 for location of bulkhead). This cable may be discarded.

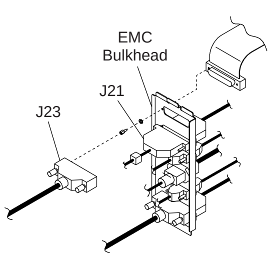


Figure 9 Side Bulkhead

4.) Disconnect the cables from both sides of the side bulkhead at J21 (use the stand-off tool). The cable from the Octane side is now free to be discarded.

⚠ **NOTICE**
Exercise finesse to remove J21

Use of the supplied AMP tool is recommended. Failure to use an appropriate stand-off tool to remove the two nuts from J21 will result in the press-fit nuts breaking loose from the inside of the cable, complicating the removal of the cable. DO NOT press on the handle of the AMP tool when removing the nuts. Be gentle and exercise finesse.

5.) Remove the Intercom/Interconnect cable by doing the following:

a.) Disconnect the RJ45 cable from the “Debug” port of the Motorola board (see Figure 10).

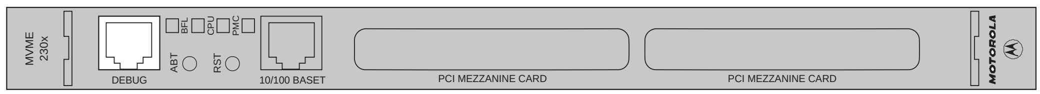


Figure 10 Motorola Board (end view)

b.) Using a small flat-blade screwdriver, disconnect the J4 connector from the Intercom Assembly (see Figure 11).

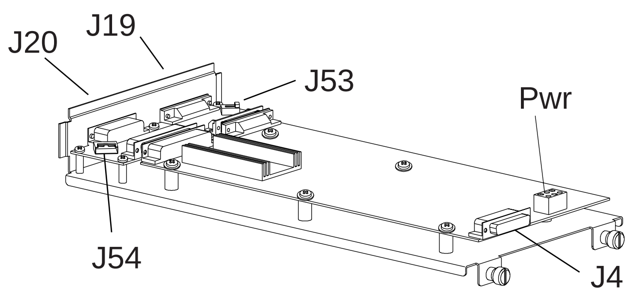


Figure 11 Intercom Assembly

Note: It may be necessary to remove the intercom board’s power cable before continuing.

c.) Loosen the two captive screws and slide the assembly out

d.) Disconnect J54 from the Remote Tilt board, by depressing clip.

e.) Discard the old Intercom/Interconnect cable.

6.) Connect the Intercom/Interconnect octopus (2269875-2) to J21, using the screw-locks.

7.) Install the new Intercom/Interconnect octopus (2269875-2) by doing the following:

a.) Connect J54 to the Remote Tilt board.

b.) Slide the assembly board into place and tighten the two captive screws. Make sure that the fiber-optic cable is well out of the way of the board as it is slid into place.

c.) Reattach the power cable, if necessary.

d.) Connect the J4 connector to the Intercom Assembly.

⚠ **NOTICE**
Do not over-tighten thumbscrews.

When tightening the thumbscrews on the J4 connector, simply snug the screws with your fingers. Do not over-tighten.

e.) Connect the RJ45 cable to the “Debug” port of the Motorola board (see Figure 10).

f.) Neatly coil and tie-up the unused Pegasus cable.

8.) Install the new serial octopus (2269874-2) to the Octane side of J21.

9.) Connect the new SCSI cable (2277402) to J23.

10.) Connect the new SCSI cable (2274402)—from J23—to the Octane’s SCSI port, freed in step 7.

4 DASM Connection & Settings

1.) If your system uses a DASM, remove the existing SCSI cable from the DASM.
2.) Connect the new cable (2277403) between the newly installed SCSI card and the DASM.
3.) Set the termination switch on the bottom of the DASM to the ON position (see Figure 12).

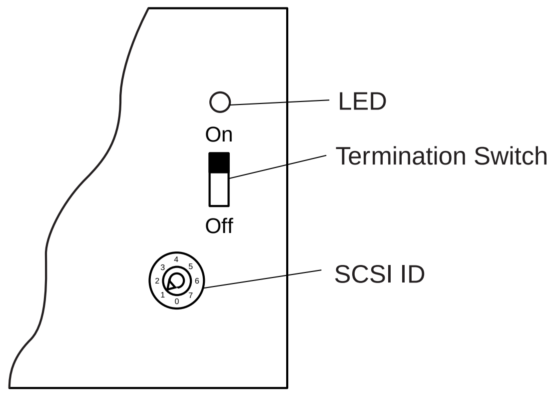


Figure 12 Cutaway of DASM bottom, showing Termination Switch set to ON

5 Image Generator Upgrade (if applicable)

Perform the Pegasus Image Generator upgrade (FMI 2291959), now, if applicable.

6 Console Reassembly

1.) Replace the "ICEbox" EMC cover.
2.) Return the Octane to its original position.
3.) Replace the console's front and rear covers.
4.) Reattach connectors J19, J20 and J52 to the rear bulkhead.
5.) Verify that no personnel are in a potentially hazardous position, and restore power to the system.

7 Load Software

When this SCSI Upgrade FMI and the Pegasus Image Generator Upgrade FMI (if applicable) are complete, proceed to the 2.1 Software Upgrade, FMI25277. DO NOT load software until all hardware changes are complete.

Console Block Diagram

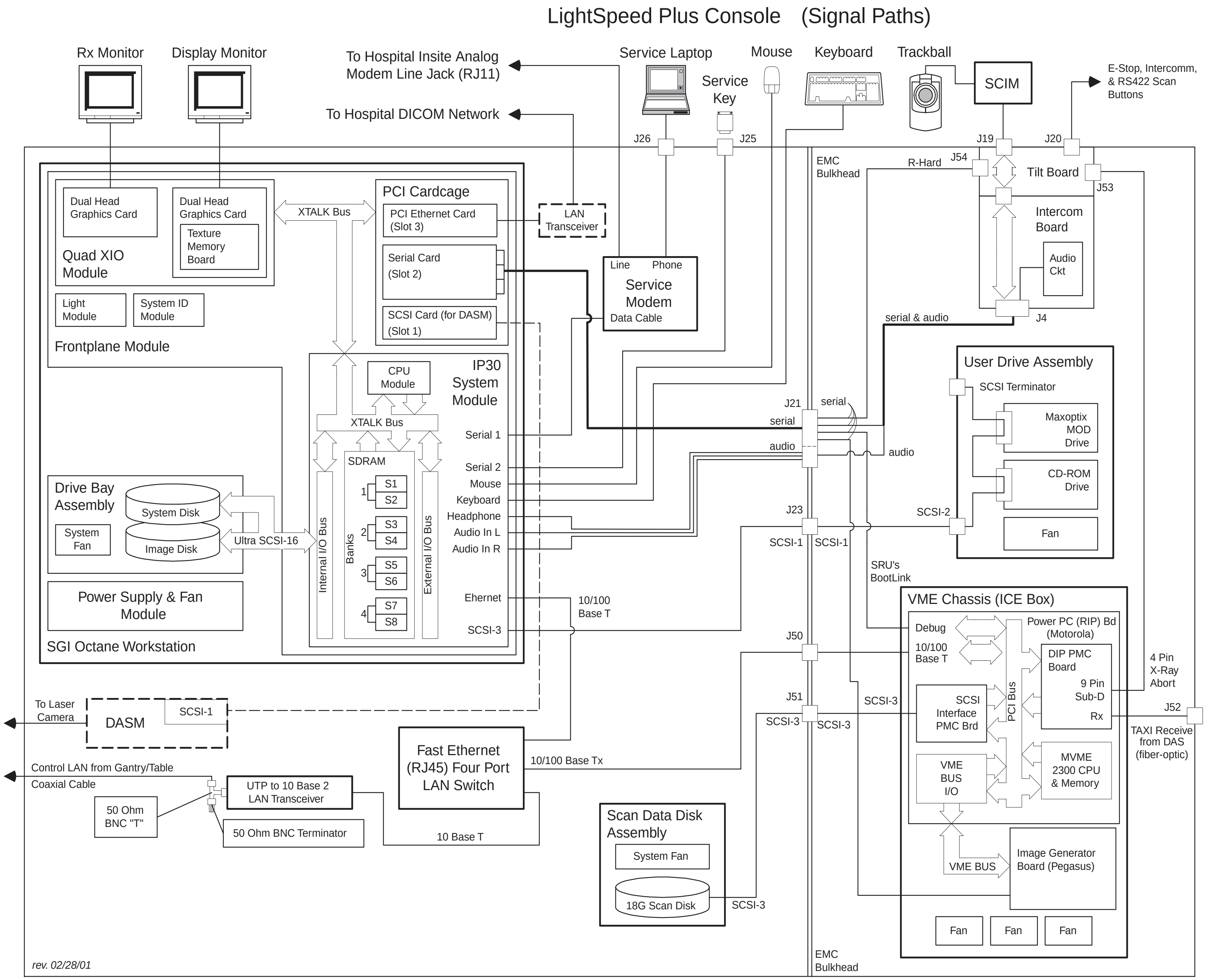


Figure 13 Console Block Diagram

Troubleshooting

1 Checking the HW and SW Installation

In the unlikely event you encounter a problem, the following checks can be preformed to help isolate the source of the failure. The checks are simple. First, verify the hardware is recognized. Next verify that its software driver is loaded into memory correctly. Finally, make sure the device is mapped properly by the operating system to the device. To check that this has been done, do the following using pipe (**|**), **`grep`** and **`tail`** UNIX commands:

1.) Open an Unix shell.

2.) Check that the Serial hardware is recognized, by using the **`hinv`** command.

   a.) At the prompt, type: **`hinv | grep PCI`**

   b.) If the Serial card is detected, you will see the following PCI card listed:
   `PCI card, bus 0, slot 2, Vendor 0x114f, Device 0x4`
   Verify the vendor ID for the serial card listed is `0x114f`

3.) Check that SCSI cards are recognized, by using the **`hinv`** command.

   a.) At the prompt, type: **`hinv | grep Integral`**

```
Integral SCSI controller 0: Version QL1040B (rev. 2), single ended
Integral SCSI controller 1: Version QL1040B (rev. 2), single ended
Integral SCSI controller 2: Version QL1040B (rev. 2), single ended
Integral Fast Ethernet: ef0, version 1, pci 2
```

   b.) Inspect the screen output. Three SCSI controllers should be listed.
   `Controller 0:` -> SCSI card for disk drives
   `Controller 1:` -> SCSI card for CD-ROM and MOD
   `Controller 2:` -> SCSI card for DASM

4.) Verify the software driver for the Serial card is in memory, by using the **`showprods`** command.

   a.) At the prompt, type: **`showprods | grep cdp`**

   You should see the following, if you have installed version 1.0 serial software drivers.

```
I cdp               03/09/2001 Digi ClassicBoard PCI Adapters
I cdp.man           03/09/2001 Digi ClassicBoard PCI Documentation
I cdp.man.relnotes  03/09/2001 Digi ClassicBoard PCI ReleaseNotes
I cdp.sw            03/09/2001 Digi ClassicBoard PCI Software
I cdp.sw.base       03/09/2001 Digi ClassicBoard PCI Base Software
```

   You should see the following, if you have installed version 1.1 serial software drivers.

```
I cdpci                03/09/2001 Digi ClassicBoard PCI Adapters
I cdpci.man            03/09/2001 Digi ClassicBoard PCI Documentation
I cdpci.man.relnotes   03/09/2001 Digi ClassicBoard PCI ReleaseNotes
I cdpci.sw             03/09/2001 Digi ClassicBoard PCI Software
I cdpci.sw.base        03/09/2001 Digi ClassicBoard PCI Base Software
```

5.) Software support for SCSI cards is embedded automatically in the IRIX OS kernel and cannot be viewed.

6.) Inspect the `/etc/uucp/Devices` file and verify the OS recognizes the serial devices.

a.) At the prompt, type: **`cd /etc/uucp`**

b.) Now type: **`tail -5 Devices`**

The **`tail`** command lists the last 5 lines in the ASCII file called `Devices`.

```
# Serial line connection for the sbc
ICE ttydp02 - 9600 direct
rhard ttydp00 - 9600 direct
pig ttydp01 - 57600 direct
ACU /dev/ttyf1 null 38400 212 x MultiTech
```

c.) Inspect the output. If the serial card is mapped correctly, you will see CT serial connections listed, such as `ICE`, `rhard`, `pig` and `ACU` hardware. The last line (above) will only appear if InSite Software has been loaded.

7.) Use **`scsistat`** to list attached SCSI devices recognized by the OS.

Note: DASM must be attached and "ON" to check PCI SCSI Card

If no device (i.e. DASM) is attached to the PCI SCSI card's external connector, the PCI SCSI card is not listed in the scsistat output.

a.) At the prompt, type: **`scsistat`**

```
Device 0 1  Disk      SGI      IBM DNES-309170Y FW Rev: SA30
Device 0 2  Disk      SGI      IBM DNES-309170Y FW Rev: SA30
Device 1 3  Optical   Maxoptix T5-2600          FW Rev: H4.2
Device 1 6  CD-ROM    TEAC     CD-ROM CD-532S   FW Rev: 1.0A
Device 2 1  Disk      CDA      DASM-VDB         FW Rev: 3.0A
```

b.) Inspect the screen output and verify that the DASM is recognized.

2 Corrective Actions

Depending on the check that failed above, choose an action that's appropriate.

1.) Card not detected by the **`hinv`** command. Check connections and replace the board. Printed circuit cards are not field repairable.

2.) Output listing of **`showprods`** command is incorrect. The correct serial drivers must be installed for the device to function properly.

- If `cdp` or `cdpci` cannot be found using the **`showprods`** command, the drivers have not been installed. Re-boot system and re-load system software if necessary.
- If **`showprods`** lists `cdp` as a driver filename and the OS installed is version 6.5.9 or higher, the serial drivers will not function properly. Using a cdp driver with 6.5.9 or higher OS will cause the computer to "hardlock", upon using the PCI serial ports. Re-boot system and re-load system software if necessary.

3.) No serial hardware listed in the Devices file. If `ICE`, `rhard`, `pig` and `ACU` hardware is not listed in the `Devices` file, re-boot and re-load system software if necessary.

End of Document